Free balancing equations worksheet worksheet 10

Maths | 7th grade | Algebra

1. If 3x + 7 = 25, what is the value of x?

A) 5
B) 6
C) 4
D) 7

2. Solve for y: 5y - 9 = 2y + 12

A) 3
B) 5
C) 7
D) 10

3. The sum of a number and twice its value is 33. What is the number?

A) 11
B) 10
C) 12
D) 15

4. What value of t makes the equation 4t + 9 = 3t + 15 true?

A) 3
B) 4
C) 6
D) 7

5. If 7 - 2n = 1, what is the value of n?

A) 3
B) 4
C) 2
D) 5

6. Solve for x: 8x + 3 = 4x + 19

A) 3
B) 4
C) 5
D) 6

7. Which of the following is a solution to the equation 6k - 4 = 2k + 20?

A) 6
B) 8
C) 7
D) 9

8. If 5(m + 2) = 35, what is the value of m?

A) 5
B) 4

C) 3
D) 2

9. What is the value of x if 7x + 12 = 40?

A) 2
B) 3
C) 4
D) 5

10. Solve for p in the equation 2p - 6 = p + 9

A) 15
B) 12
C) 10
D) 7

Answer Key

1. A
Subtract 7 from both sides to get 3x = 18, then divide by 3 to find x = 6.

2. C
Subtract 2y from both sides to get 3y - 9 = 12, then add 9 to both sides to find 3y = 21. Divide by 3 to get y = 7.

3. A
If x is the number, then x + 2x = 33. This simplifies to 3x = 33, so x = 11.

4. B
Subtract 3t from both sides to get t + 9 = 15, then subtract 9 from both sides to find t = 6.

5. C
Subtract 7 from both sides to get -2n = -6, then divide by -2 to find n = 3.

6. A
Subtract 4x from both sides to get 4x + 3 = 19, then subtract 3 from both sides to find 4x = 16. Divide by 4 to get x = 4.

7. A
Subtract 2k from both sides to get 4k - 4 = 20, then add 4 to both sides to find 4k = 24. Divide by 4 to get k = 6.

8. B
First, distribute the 5 to get 5m + 10 = 35. Subtract 10 from both sides to get 5m = 25, then divide by 5 to find m = 5.

9. C
Subtract 12 from both sides to get 7x = 28, then divide by 7 to find x = 4.

10. D
Subtract p from both sides to get p - 6 = 9, then add 6 to both sides to get p = 15.